

TREE 1

PROBLEM 10

QUESTION:

DS	Session	Tree 1	Question Information	Level 1	Challenge 70
Question description You are given a list consisting of n integers. Your task is to remove elements from the list at given positions, and report the removed elements. Constraints $1 \leq n \leq 2 \cdot 10^5$ $1 \leq x_i \leq 10^9$ $1 \leq p_i \leq n - i + 1$ Input The first input line has an integer n: the initial size of the list. During the process, the elements are numbered $1, 2, \dots, k$ where k is the current size of the list. The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the list. The last line has n integers $p_1, p_2, \dots, p_n$: the positions of the elements to be removed. Output Print the elements in the order they are removed.					

Logical Test Cases

Test Case 1 INPUT (STDIN) <pre>5 2 6 1 4 2 3 1 3 1 1</pre> EXPECTED OUTPUT <pre>1 2 2 6 4</pre>	Test Case 2 INPUT (STDIN) <pre>6 7 2 6 1 4 2 11 3 1 3 1 1</pre> EXPECTED OUTPUT <pre>2 6 7 4 2 1</pre>
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Mandatory Test Cases

Test Case 1 KEYWORD <pre>void build(int k,int l,int r)</pre>

Complexity Test Cases

Test Case 1 CYCLOMATIC COMPLEXITY 2	Test Case 2 TOKEN COUNT 330	Test Case 3 NLOC 37
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SOURCE CODE:

```
#include <stdio.h>

int main() {
    int length;
    if (scanf("%d", &length) != 1) return 0;

    int collection[length];
    for (int i = 0; i < length; i++) scanf("%d", &collection[i]);

    int current_len = length;
    for (int step = 0; step < length; step++) {
        int rank;
        scanf("%d", &rank);
        int target_pos = rank - 1;
        printf("%d%s", collection[target_pos], (step == length - 1) ? "" : " ");

        for (int j = target_pos; j < current_len - 1; j++) {
            collection[j] = collection[j + 1];
        }
        current_len--;
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

```
5
2 6 1 4 2
3 1 3 1 1
1 2 2 6 4
```

```
6
7 2 6 1 4 2
11 3 1 3 1 1
2 6 7 4 2 1
```